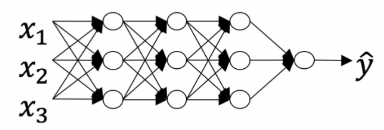
1. **Which of the following are true? (Check all that apply.)** 1 point

- [x] $a^{[2]}$ denotes the activation vector of the 2^{nd} layer.
- [x] $a^{[2](12)}$ denotes the activation vector of the 2^{nd} layer for the 12^{th} training example.
- [x] $a^{[2]}_4$ is the activation output by the 4^{th} neuron of the 2^{nd} layer
- [ ] $a^{[2]}_4$ is the activation output of the 2^{nd} layer for the 4^{th} training example
- [ ] X is a matrix in which each row is one training example.
- [x] X is a matrix in which each column is one training example.
- [ ] $a^{[2](12)}$ denotes activation vector of the 12^{th} layer on the 2^{nd} training example.

2. **In which of the following cases is the linear (identity) activation function most likely used?** 1 point

- [ ] As activation function in the hidden layers.
- [x] When working with regression problems.
- [ ] The linear activation function is never used.
- [ ] For binary classification problems.

3. **Which of the following represents the activation output of the second neuron of the third layer applied to the fourth example?** 1 point



- [x] $a_2^{[3](4)}$
- [ ] $a_2^{[4](3)}$
- [ ] $a_4^{[3](2)}$
- [ ] $a_3^{[4]2}$

4. **The use of the ReLU activation function is becoming more rare because the ReLU function has no derivative for $c = 0$. True/False?** 1 point

- [x] False
- [ ] True

5. **Consider the following code:** 1 point

#+begin_src python

x = np.random.rand(4, 5)

y = np.sum(x, axis=1)

#+end_src

What will be y.shape?

- [ ] (5,)
- [x] (4,)
- [ ] (4, 1)
- [ ] (1, 5)

6. **Suppose you have built a neural network. You decide to initialize the weights and biases to be zero. Which of the following statements is true?** 1 point

- [ ] Each neuron in the first hidden layer will perform the same computation in the first iteration. But after one iteration of gradient descent they will learn to compute different things because we have "broken symmetry".
- [x] Each neuron in the first hidden layer will perform the same computation. So even after multiple iterations of gradient descent, each neuron in the layer will be computing the same thing as other neurons.
- [ ] Each neuron in the first hidden layer will compute the same thing, but neurons in different layers will compute different things, thus we have accomplished "symmetry breaking" as described in the lecture.
- [ ] The first hidden layer's neurons will perform different computations from each other even in the first iteration; their parameters will thus keep evolving in their own way.

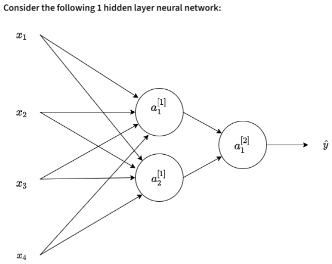
7. **A single output and single layer neural network that uses the sigmoid function as activation is equivalent to the logistic regression. True/False** 1 point

- [ ] False
- [x] True

8. **You have built a network using the tanh activation for all the hidden units. You initialize the weights to relatively large values, using np.random.randn(..,..)*1000. What will happen?** 1 point

- [ ] So long as you initialize the weights randomly gradient descent is not affected by whether the weights are large or small.
- [ ] This will cause the inputs of the tanh to also be very large, thus causing gradients to also become large. You therefore have to set α to a very small value to prevent divergence; this will slow down learning.
- [ ] This will cause the inputs of the tanh to also be very large, causing the units to be "highly activated" and thus speed up learning compared to if the weights had to start from small values.
- [x] This will cause the inputs of the tanh to also be very large, thus causing gradients to be close to zero. The optimization algorithm will thus become slow.

9. **Consider the following 1 hidden layer neural network:** 1 point

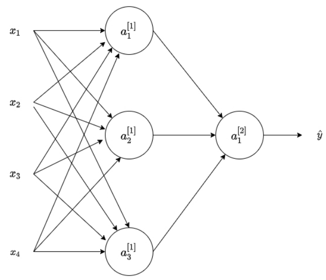


Which of the following statements are True? (Check all that apply).

- [ ] $b^{[1]}$ will have shape (4, 2)

- [x] $b^{[1]}$ will have shape (2, 1).
- [ ] $W^{[2]}$ will have shape (2, 1)
- [x] $W^{[1]}$ will have shape (2, 4).
- [x] $W^{[2]}$ will have shape (1, 2)
- [ ] $W^{[1]}$ will have shape (4, 2).

10. Consider the following 1 hidden layer neural network: 1 point



What are the dimensions of $Z^{[1]}$ and $A^{[1]}$?

- ( ) $Z^{[1]}$ and $A^{[1]}$ are (3, 1)
- ( ) $Z^{[1]}$ and $A^{[1]}$ are (4, m)
- (•) $Z^{[1]}$ and $A^{[1]}$ are (3, m)
- ( ) $Z^{[1]}$ and $A^{[1]}$ are (4, 1)